DETAILED STUDY OF THE MIRROR SYMMETRY STRUCTURE OF ^{27}AL AND ^{27}SI

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Gamow-Teller transitions

- Gamow–Teller (GT) transitions are the most common weak interaction processes of spin– isospin ($\sigma\tau$) type in atomic nuclei
- Direct beta decay studies give limited information
- Can use charge exchange reactions to access higher excitation energies
- Can compare to M1 transitions ("quasi-proportional"), CG coefficients are different → deduce T

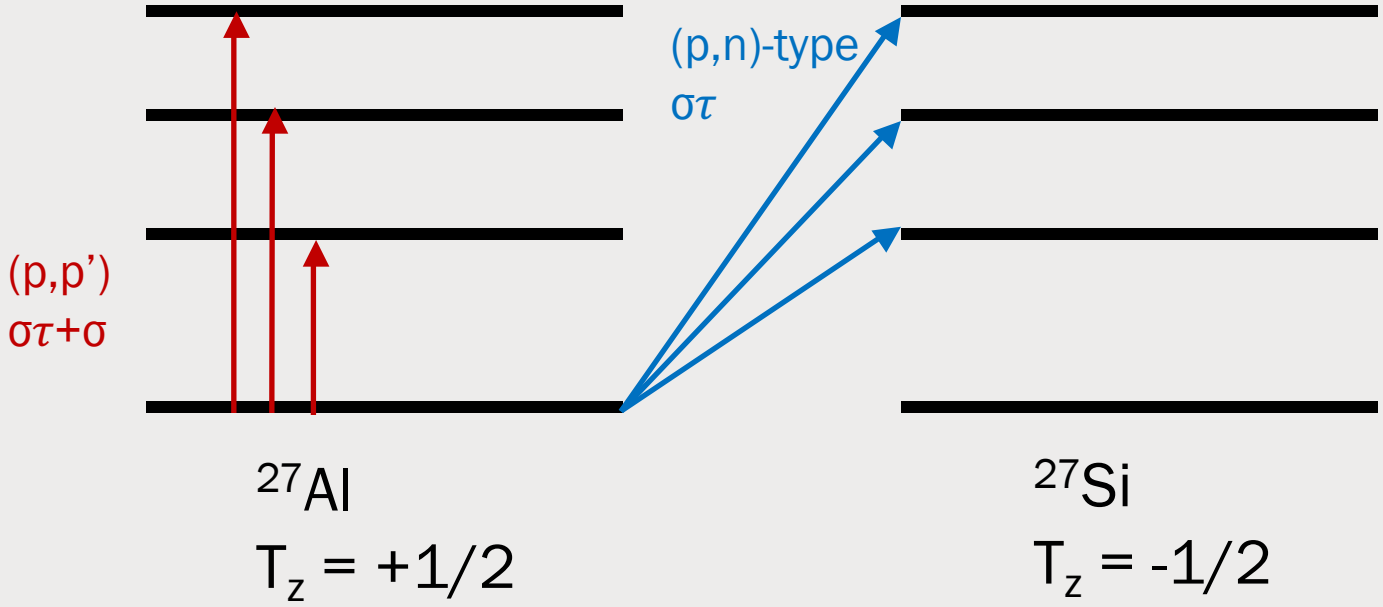
Extracting B(GT) values

- B(GT) value is proportional to the measured cross-section
- Since beta decay gives absolute B(GT) values, the B(GT) from CE can be obtained by normalizing to the known B(GT) at lower excitation energies

$$\frac{\mathrm{d}\sigma_{\mathrm{GT}}}{\mathrm{d}\Omega}(q,\omega) \simeq K(\omega)N_{\sigma\tau}|J_{\sigma\tau}(q)|^2 B(\mathrm{GT})$$
$$= \hat{\sigma}_{\mathrm{GT}}(q,\omega)B(\mathrm{GT}),$$

$$\frac{\mathrm{d}\sigma_{M1\sigma}}{\mathrm{d}\Omega}(q,\omega) \simeq K(\omega)N_{\sigma\tau}|J_{\sigma\tau}(q)|^2 B(M1_\sigma)$$
$$= \hat{\sigma}_{M1\sigma}(q,\omega)B(M1_\sigma),$$

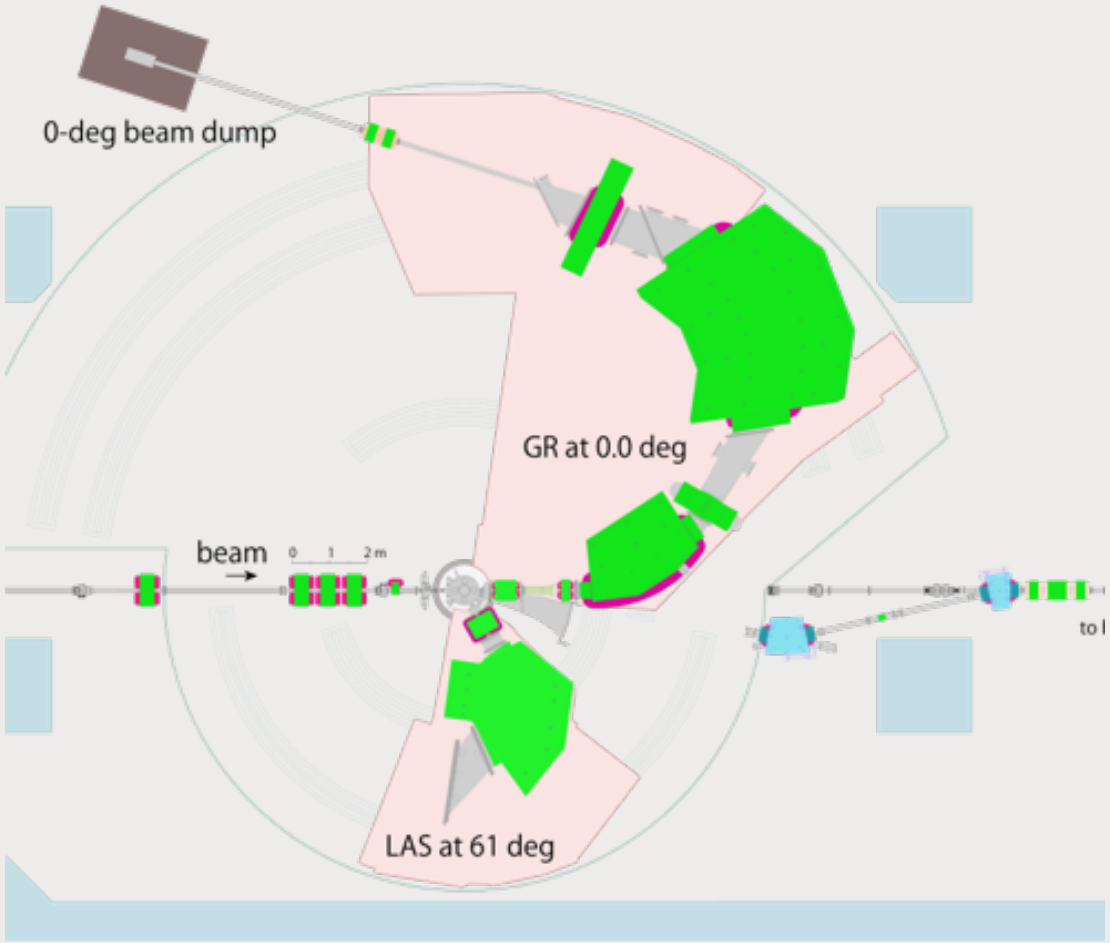
Isospin symmetry in the mirror nuclei



- Mirror symmetry in ^{27}Al and ^{27}Si was studied with ^{27}Al(^{3}He,t)^{27}Si and ^{27}Al(p,p') reactions

Measurement approach

- Measurements conducted at RCNP, Osaka University facility using AVF and RING cyclotrons, using the Grand Raiden spectrometer at the WS Experimental Hall



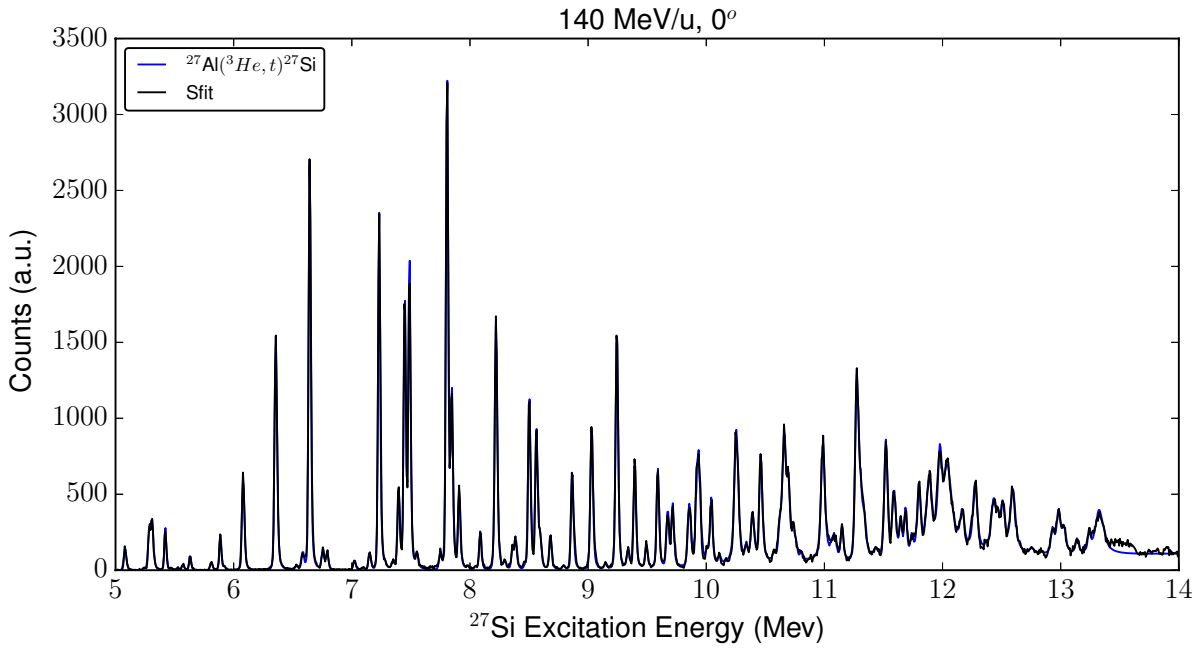
^{27}Al(^{3}He,t) at 0°, 140 MeV/u
^{27}Al(p,p') at 0°, 295 MeV

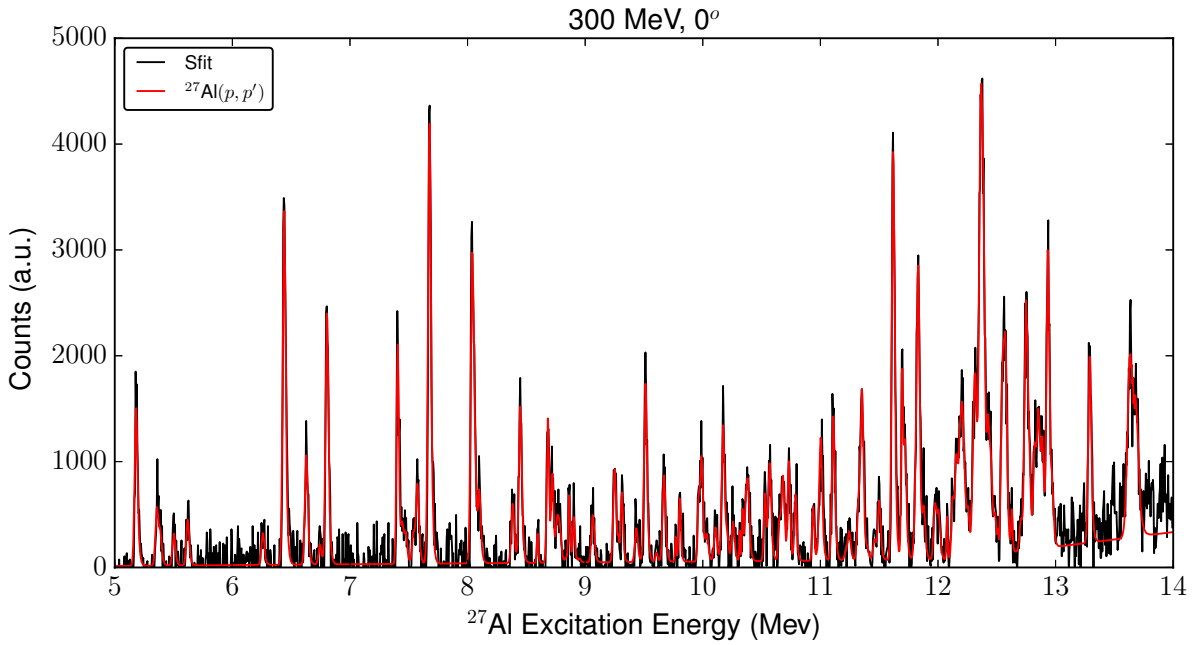
Excitation-energy resolution of 35 keV achievable using the dispersion matching technique

D1 magnet stops the ^{3}He, only see tritons

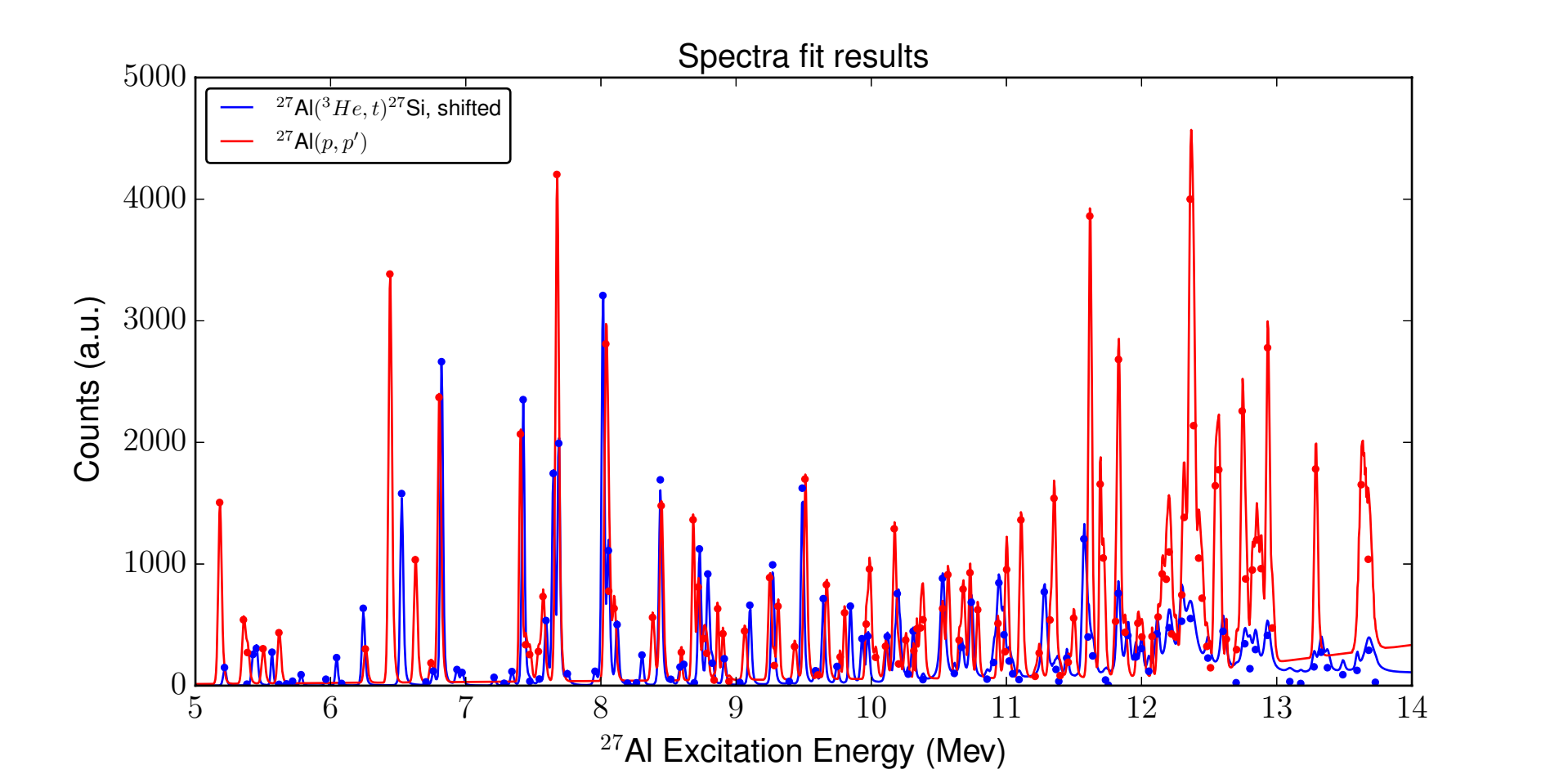
2 VDC's were used for position detection
2 Plastic scintillators for particle ID and trigger

Comparing this to the work done by Fujita (Phys.Rev. C **59**, 90 (1999)), with improved statistics and resolution

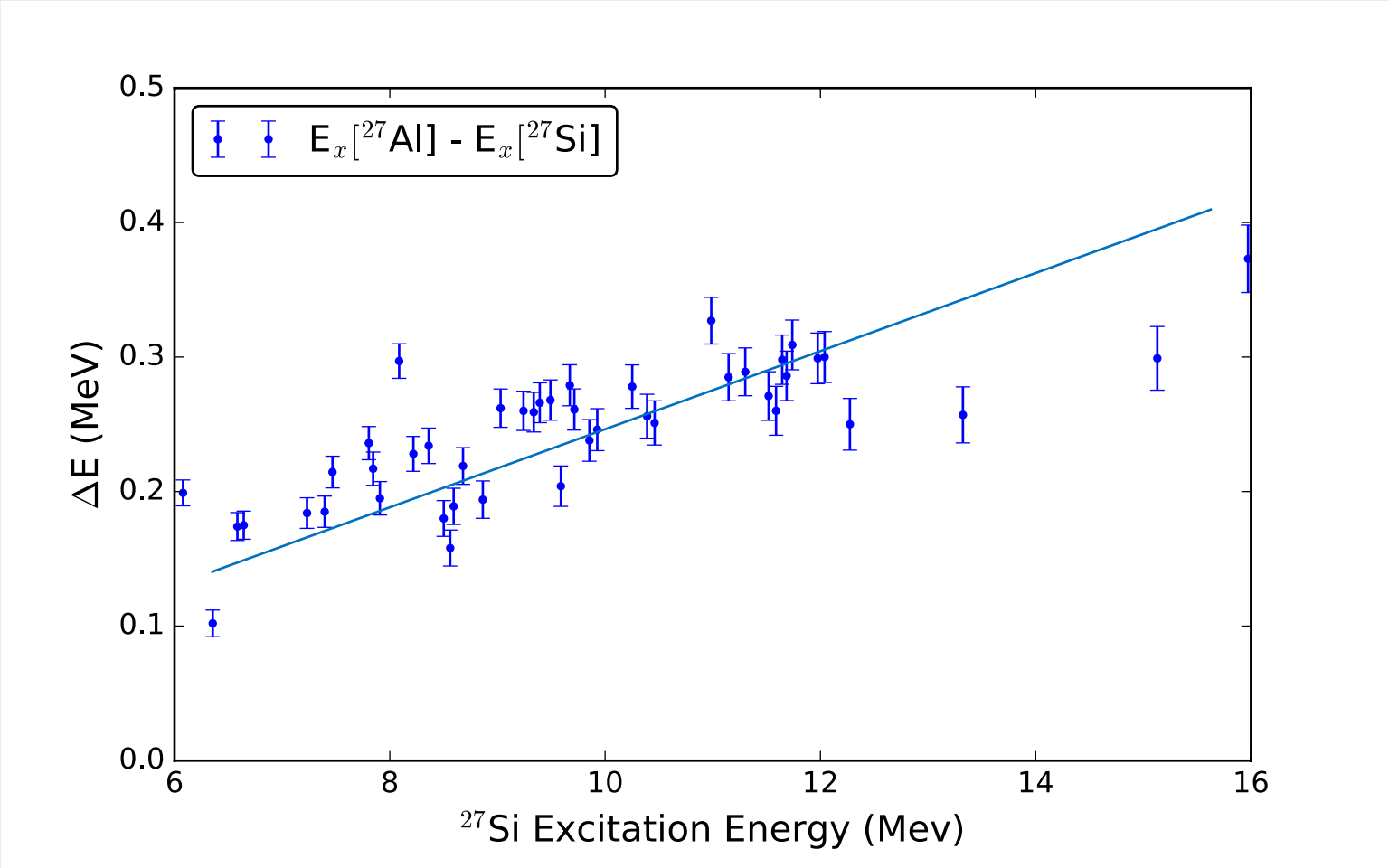

140 MeV/u, 0°
^{27}Al($^3He,t$)^{27}Si
Sfit
Counts (a.u.)
^{27}Si Excitation Energy (Mev)


300 MeV, 0°
Sfit
^{27}Al(p,p')
Counts (a.u.)
^{27}Al Excitation Energy (Mev)

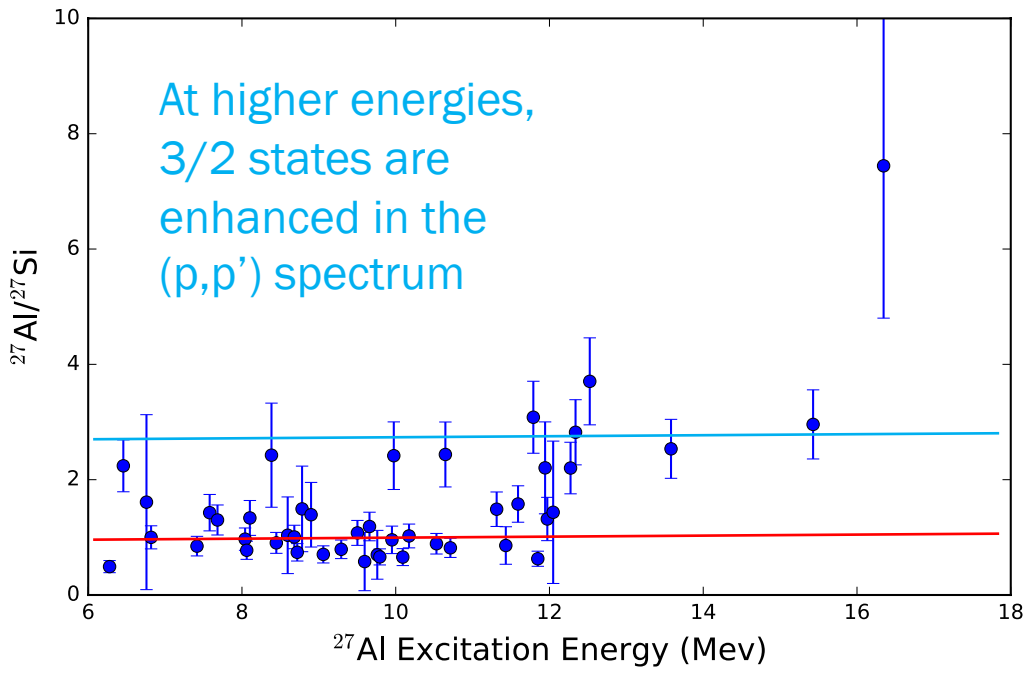
Mirror symmetry

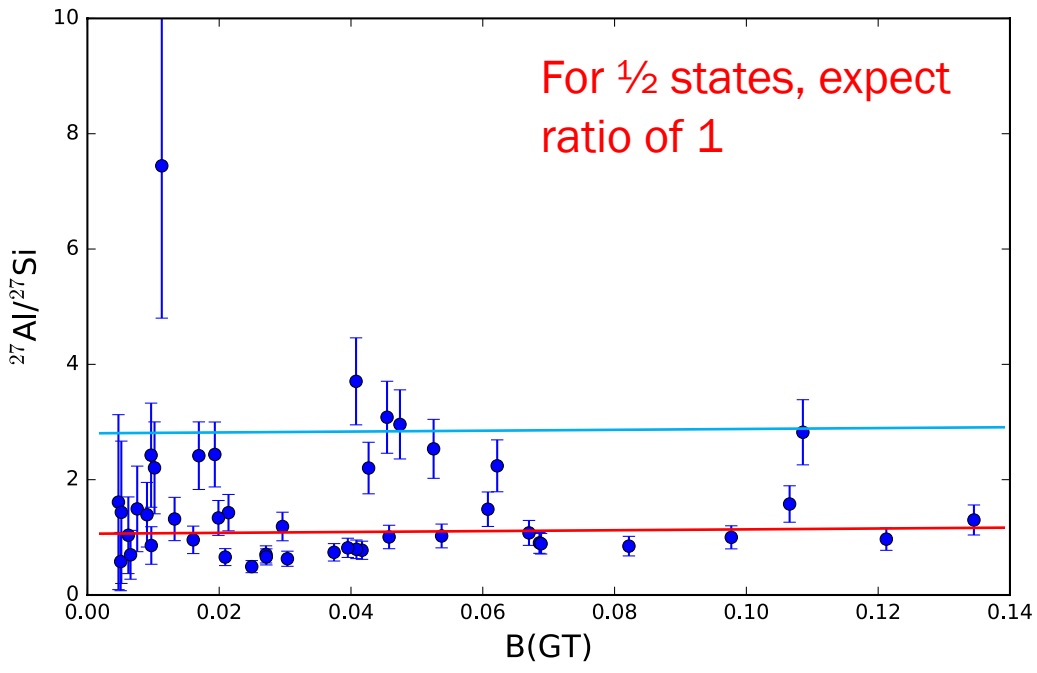


Energy shift between analogous states

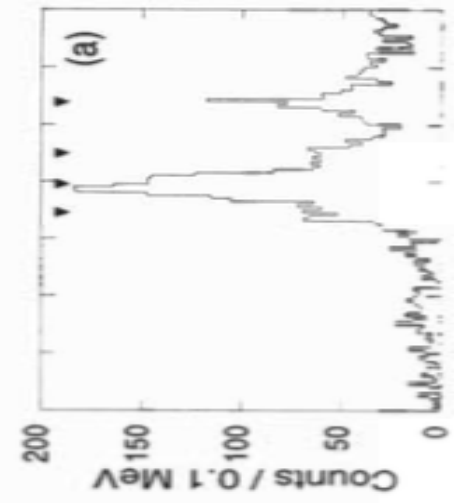


Ratio of ^{27}Al to ^{27}Si peaks (corrected and normalized)

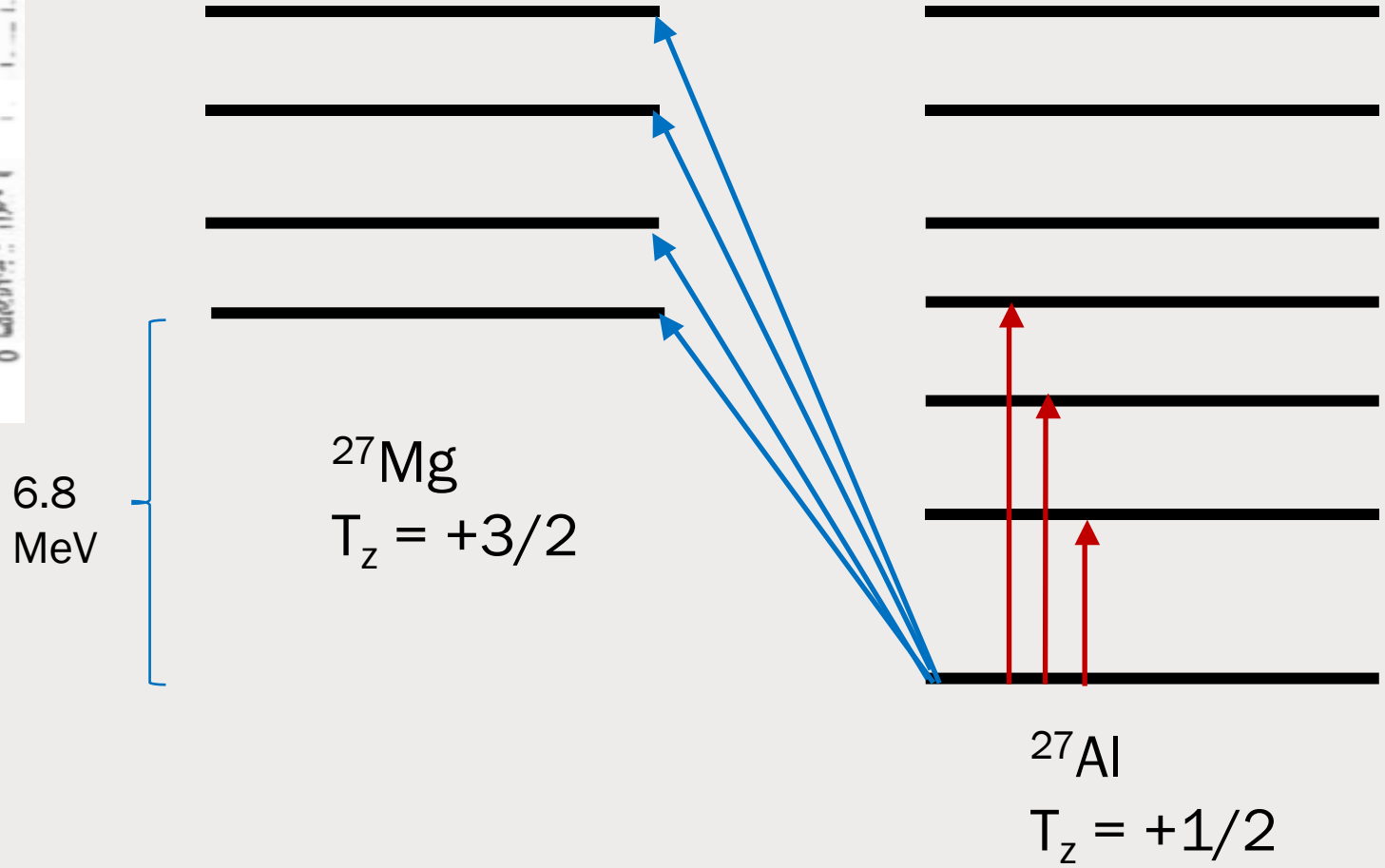




$^{27}Al(d,^{2}He)\,^{27}Mg$: only T=3/2

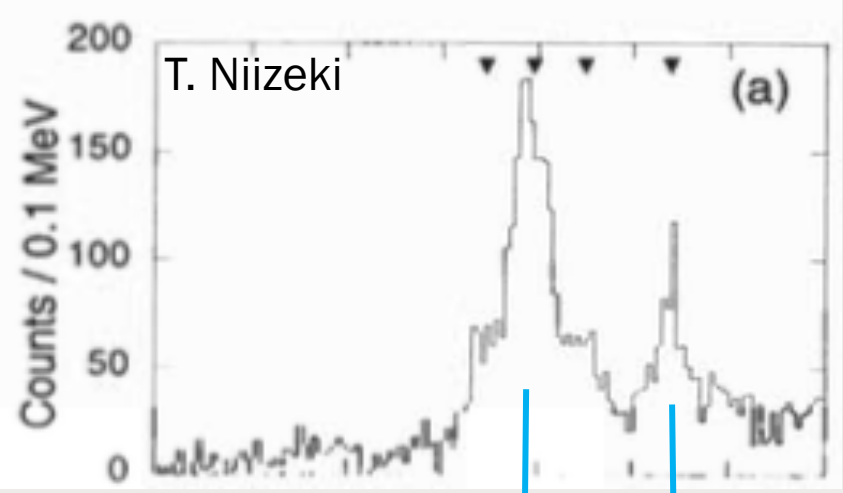


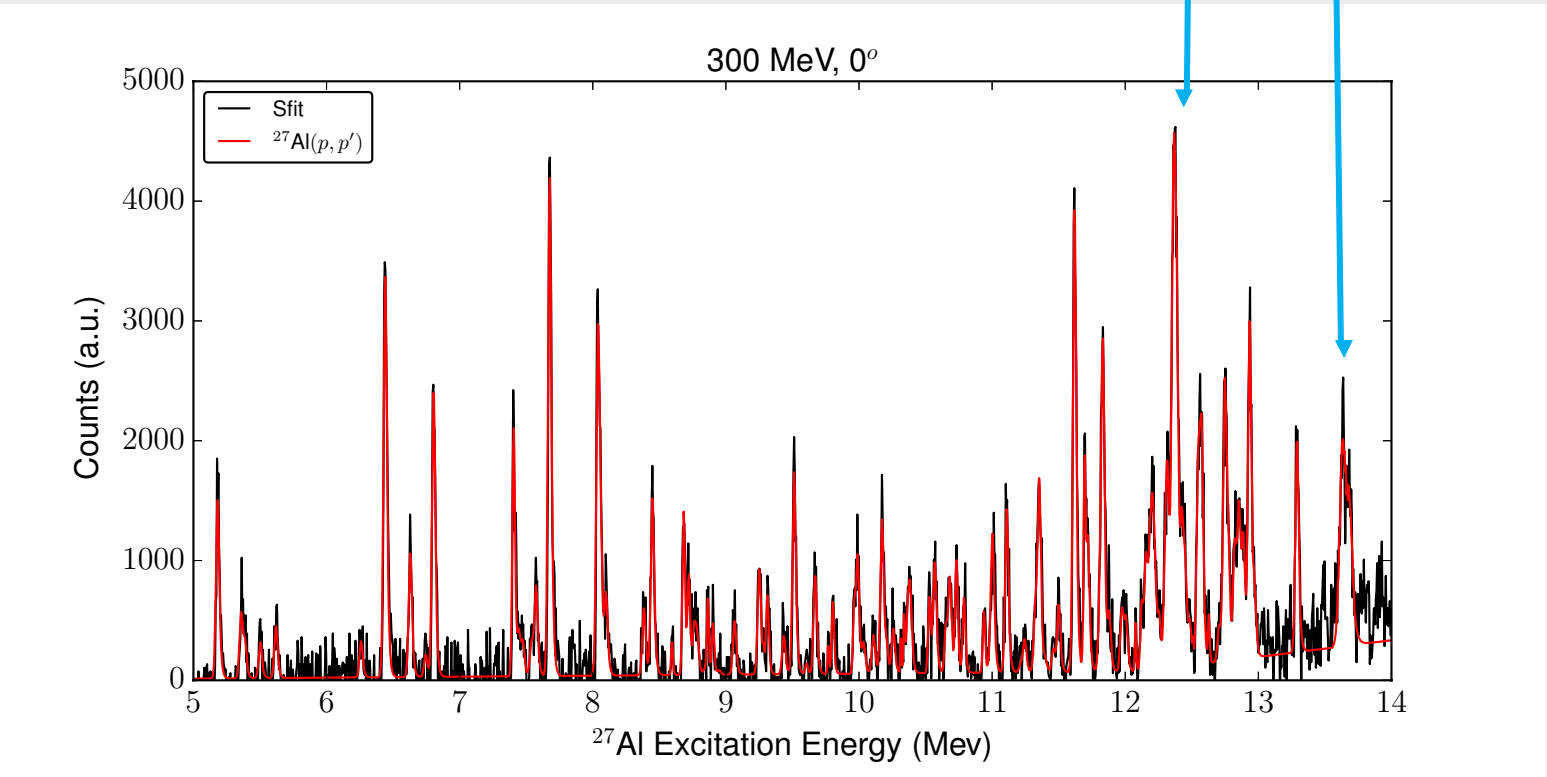
T. Niizeki



Spin comparison

^{27}Mg states correspond to the enhanced states with Al/Si ratio of ~3 in the analysis





Conclusion

- Deduced transitions to 3/2 isospin states from enhancement in M1 (should be x4 based on CG coefficients) and from comparison to ^{27}Mg data
- Extended and corrected B(GT) values
- Extended analogous states